MATH 401. Homework 4. Sadullaev L.

Problem 1.

Assume that $\{x_n\}$, $\{y_n\}$, $\{z_n\}$ are three sequences s.t. $x_n \leq y_n \leq z_n$ $\forall n \in \mathbb{N}$. Use the definition of convergence to show that if $\{x_n\}$, $\{z_n\}$ both converge to a, then $\{y_n\}$ also converges to a. (this is sometimes called squeeze theorem)

Solution:

Let x_n, y_n, z_n be the sequences s.t.

(1) $x_n \leq y_n \leq z_n \quad \forall n \in \mathbb{N}$

Assume $\{x_n\} \to a$ & $\{z_n\} \to a$.

WTS (want to show) $\{y_n\} \to a$.

(2) $\{x_n\} \to a \Rightarrow \forall \varepsilon, \exists N_1 \in \mathbb{N}$ s.t.

$$|x_n - a| < \varepsilon \quad \forall n \geq N_1.$$

$\{z_n\} \to a \Rightarrow \forall \varepsilon, \exists N_2 \in \mathbb{N}$ s.t.

(3) $|z_n - a| < \varepsilon \quad \forall n \geq N_2.$

Using (1), (2), and (3),

(4) let $N = \max\{N_1, N_2\}$ and by (1)

$$x_n \le y_n \le z_n \quad \forall n \in \mathbb{N}$$

subtract a $\Rightarrow$

(5) $$x_n - a \le y_n - a \le z_n - a$$

$$|x_n - a| < \varepsilon \Rightarrow \overbrace{-\varepsilon < x_n - a < \varepsilon}^{(6)} \quad \forall n \ge N_1$$

$$|z_n - a| < \varepsilon \Rightarrow -\varepsilon < \underbrace{z_n - a < \varepsilon}_{(7)} \quad \forall n \ge N_2$$

Applying the underlined parts (6) & (7) for (5):

(8) $$\underline{\underline{-\varepsilon <}}\ x_n - a \le \underline{\underline{y_n - a}} \le z_n - a\ \underline{\underline{< \varepsilon}}$$

$\forall n \ge N = \max\{N_1, N_2\}$ look at (4) above.

(8) $\Rightarrow$ $-\varepsilon < y_n - a < \varepsilon \quad \forall n \ge N$

$\Rightarrow |y_n - a| < \varepsilon \quad \forall n \ge N \Rightarrow$

$\{y_n\} \to a$.

$\blacksquare$

Problem 2.1.3.

Use the definition of continuity to show that if $f: \mathbb{R} \to \mathbb{R}$ is continuous at a, then the function $g(x) = M f(x)$, where $M = \text{const}$, is also constinuos at a.

Solution:

Let $f: \mathbb{R} \to \mathbb{R}$ be a continuous function at a.

Consider the definition of continuity at a point:

(1) $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $|f(x) - f(a)| < \varepsilon$ whenever $|x-a| < \delta$.

Consider the function $g(x) = M \cdot f(x)$, $M = \text{const}$.

WTS:

(2) $\forall \varepsilon > 0, \exists \delta_1 > 0$ s.t. $|g(x) - g(a)| < \varepsilon$ whenever $|x-a| < \delta_1$.

(3) $|g(x) - g(a)| = |Mf(x) - Mf(a)| = |M| \cdot |f(x) - f(a)|$

Same epsilon

But for $\varepsilon' = \dfrac{\varepsilon}{|M|}$ $\exists \delta_2 > 0$ s.t.

(4) $|f(x) - f(a)| < \varepsilon' = \dfrac{\varepsilon}{|M|}$ when $|x-a| < \delta_2$.

Let $\delta_1 = \delta_2$ (5)

then

$$|g(x) - g(a)| = |M|\,|f(x) - f(a)| <$$

$$< \cancel{|M|} \cdot \frac{\varepsilon}{\cancel{|M|}} = \varepsilon$$

whenever $|x - a| < \delta_1 \Rightarrow$

$g(x)$ is continuous at $x = a$.

$\blacksquare$

Problem 2.1.5.

(a) Use the definition of continuity to show that if f, g are continuous at a, so is $f \cdot g$.
(Hint: Write $|f(x) \cdot g(x) - f(a) \cdot g(a)| =$

$= |f(x) \cdot g(x) - f(a) \cdot g(x) + f(a) \cdot g(x) - f(a) \cdot g(a)|$ and

use the Triangle Inequality.)

(b) Combine the result from part (a) with the proposition 2.1.9. to show that if f and g are continuous at a, and $g(a) \neq 0$, then $\frac{f}{g}$ is continuous at a.

Solution:

Let f, g be continuos functions from $\mathbb{R}$ to $\mathbb{R}$.

WTS: $f \cdot g$ is con-s as well $\Rightarrow$ $\forall \varepsilon > 0, \exists \delta > 0 \ni$

(1) $|f(x) \cdot g(x) - f(a) g(a)| < \varepsilon$ whenever $|x-a| < \delta. \Rightarrow$

$$(2) \Rightarrow |f(x) \cdot g(x) - f(a) \cdot g(x) + f(a) \cdot g(x) - f(a) \cdot g(a)| \leq$$

$$\leq |g(x)| \cdot |f(x) - f(a)| + |f(a)| \cdot |g(x) - g(a)|.$$

We have 4 different cases here for the formula (2) which get different scenario, let me consider them all:

Case 1: $f(a) \neq 0 \quad g(a) \neq 0.$ (3)

First let me bound $g(x)$. Since $g(x)$ is con-s at $x = a$, $\exists\, \delta_1 > 0$ s.t

(4) $$|g(x)| \leq \underbrace{2 \cdot |g(a)|}_{\text{serves as } \varepsilon_1} \quad \text{whenever} \quad |x-a| < \delta_1$$

$f(x)$ is continuous $\Rightarrow$

$$\forall\, \varepsilon_2 = \frac{\varepsilon}{2 \cdot 2 \cdot |g(a)|}, \quad \exists\, \delta_2 > 0 \text{ s.t.}$$

(5) $$|f(x) - f(a)| < \varepsilon_2 = \frac{\varepsilon}{2 \cdot 2 \underbrace{|g(a)|}_{\neq 0 \text{ by } (3)}} \quad \text{whenever} \quad |x-a| < \delta_2$$

$g(x)$ is con-s $\Rightarrow$

$$\forall\, \varepsilon_3 = \frac{\varepsilon}{2 \cdot \underbrace{|f(a)|}_{\neq 0 \text{ by } (3)}}, \quad \exists\, \delta_3 > 0 \text{ s.t.}$$

(6) $$|g(x) - g(a)| < \frac{\varepsilon}{2 \cdot |f(a)|} \quad \text{whenever} \quad |x-a| < \delta_3$$

Substituting (4), (5), (6) into (2) yields

if $\delta = \min\{\delta_1, \delta_2, \delta_3\}$ $\Rightarrow$

(7) $|f(x)\cdot g(x) - f(a)\cdot g(a)| \le \overset{\text{by (4)}}{|g(x)|}\cdot \overset{\text{by (5)}}{|f(x)-f(a)|} + |f(a)|\cdot \overset{\text{by (6)}}{|g(x)-g(a)|} <$

$$< \cancel{2\cdot|g(a)|}\cdot \frac{\varepsilon}{2\cdot \cancel{2|g(a)|}} + \cancel{|f(a)|}\cdot \frac{\varepsilon}{2\cdot \cancel{|f(a)|}} = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

(8) $\Rightarrow |f(x)\cdot g(x) - f(a)\cdot g(a)| < \varepsilon$ whenever $|x-a| < \delta$

(9) where $\delta = \min\{\delta_1, \delta_2, \delta_3\}$. $\Rightarrow$

$f(x)\cdot g(x)$ is con-s.

$\boxtimes$

Case 2: $f(a) \neq 0$ (10),
$g(a) = 0$ (11)

We need to do all previous business by considering (10) & (11).

$f(x)$ is con-s $\Rightarrow$
$\forall \varepsilon_1 = \varepsilon\cdot|f(a)|$, $\exists\, \delta_1 > 0$ s.t.

(12) $|f(x) - f(a)| = \varepsilon_1 = \varepsilon\cdot|f(a)|$ whenever $|x-a| < \delta_1$

Actually it's very easy for $f(x)$ to find δ_1 in this case.
You will find out why I exactly picked this $\varepsilon_1 = \varepsilon|f(a)|$!

$g(x)$ is continuous $\Rightarrow$

$$\forall \varepsilon_2 = \frac{\varepsilon}{2 \cdot \underbrace{|f(a)|}_{\neq 0 \text{ by } (10)}}, \quad \exists\ \delta_2 > 0 \quad \text{s.t}$$

$$(13) \quad |g(x) - \underbrace{g(a)}_{=0 \text{ by } (11)}| = |g(x) - 0| = |g(x)| < \varepsilon_2 = \frac{\varepsilon}{2|f(a)|}$$

whenever $|x - a| < \delta_2$.

Now let me bound $|g(x)|$ by using previous results (13) which makes our job easy.

If we closely look at (13) $|g(x)|$ is already bounded:

$$(14) \quad |g(x)| < \frac{\varepsilon}{2|f(a)|} < \frac{1}{2|f(a)|} \quad \text{whenever } |x-a| < \delta_2$$

Again, substituting (12), (13), (14) into (2), and considering $\delta = \min\{\delta_1, \delta_2\}$, yields

$$(15)\quad |f(x)\cdot g(x) - f(a)\cdot g(a)| \le \overset{\text{by }(14)}{|g(x)|}\cdot \overset{\text{by }(12)}{|f(x)-f(a)|} + |f(a)|\cdot \overset{\text{by }(13)}{|g(x)-g(a)|} <$$

$$< \frac{1}{2\cdot \cancel{|f(a)|}}\cdot \varepsilon\cdot \cancel{|f(a)|} + \cancel{|f(a)|}\cdot \frac{\varepsilon}{2\cancel{|f(a)|}} = \frac{\varepsilon}{2}+\frac{\varepsilon}{2} = \varepsilon$$

$(16) \Rightarrow |f(x)\cdot g(x) - f(a)\cdot g(a)| < \varepsilon$ whenever $|x-a| < \delta$

(17) where $\delta = \min\{\delta_1, \delta_2\}$. $\Rightarrow$

$f(x)\cdot g(x)$ is con-s.

$\blacksquare$

Case 3: $f(a) = 0$ (18)

$g(a) \neq 0$ (19)

Now I've found an easier approach by looking at( )

WTS: (2) $|f(x)\cdot g(x) - \underbrace{f(a)}_{0}\cdot g(a)| = |f(x)\cdot g(x)| < \varepsilon$

$\forall \varepsilon > 0$.

So, $\forall \varepsilon > 0$

$$(20)\quad |f(x)\cdot g(x) - f(a)\cdot g(a)| = |f(x)\cdot g(x)| = |f(x)|\cdot|g(x)|$$

First, let me bound $|g(x)|$ by using its continuity:

$g(x)$ con. $\Rightarrow$

(21) $\exists\, \delta_1 > 0$ s.t. $|g(x)| < 2 \cdot |g(a)|$

whenever $|x-a| < \delta_1$. (Actually, it's another continuity business here!)

$f(x)$ is con. $\Rightarrow$

For $\varepsilon_2 = \dfrac{\varepsilon}{2 \cdot |g(a)|}$, $\exists\, \delta_2 > 0$ s.t

(22) $|f(x) - f(a)| = |f(x) - 0| = |f(x)| < \varepsilon_2 = \dfrac{\varepsilon}{2 \cdot |g(a)|}$

whenever $|x-a| < \delta_2$

Substituting (21) & (22) into (20) by considering $\delta = \min\{\delta_1, \delta_2\}$ yields

$$|f(x) \cdot g(x) - \overbrace{f(a)}^{=0} g(a)| = \overset{\text{by (22)}}{|f(x)|} \cdot \overset{\text{by (21)}}{|g(x)|} < \frac{\varepsilon}{\cancel{2 \cdot |g(a)|}} \cdot \cancel{2 \cdot |g(a)|} = \varepsilon.$$

(23) $\Rightarrow |f(x) \cdot g(x) - f(a) \cdot g(a)| < \varepsilon$ whenever $|x-a| < \delta$

where $\delta = \min\{\delta_1, \delta_2\}$

$\boxtimes$

Case 4: $f(a) = 0$ (24)

$g(a) = 0$ (25)

Coming back to (2) again $\forall \varepsilon > 0$, WTS

(26) $\left| f(x) \cdot g(x) - \underbrace{f(a)}_{0} \cdot \underbrace{g(a)}_{0} \right| = |f(x)| \cdot |g(x)| < \varepsilon$

for some $\delta > 0$.

in all cases same ε.

f is con-s $\Rightarrow$

$\varepsilon_1 = \sqrt{\varepsilon}$, $\exists \delta_1 > 0$ s.t.

(27) $|f(x) - f(a)| = |f(x) - 0| = |f(x)| < \varepsilon_1 = \sqrt{\varepsilon}$

whenever $|x - a| < \delta_1$

g is con-s $\Rightarrow$

For $\varepsilon_2 = \sqrt{\varepsilon}$, $\exists \delta_2 > 0$ s.t.

(28) $|g(x) - g(a)| = |g(x) - 0| = |g(x)| < \varepsilon_2 = \sqrt{\varepsilon}$

whenever $|x - a| < \delta_2$

substituting (27), (28) by $\delta = \min\{\delta_1, \delta_2\}$ into (26)

(29) $|f(x) \cdot g(x) - f(a) \cdot g(a)| = |f(x)| \cdot |g(x)| < \sqrt{\varepsilon} \cdot \sqrt{\varepsilon} = \varepsilon$

whenever $|x - a| < \delta$.

By considering all (9), (16), (23), and (29), $f(x) \cdot g(x)$ is con-s. ∎

(b) Using the result from part (a) & Prop. 2.1.9

WTS: $\frac{f(x)}{g(x)}$ is con-s.

By Prop. 2.1.9 $\frac{1}{g(x)}$ is continuous whenever $g(x)$ is continuous at a. $\Rightarrow$

(30) So $\frac{1}{g(x)}$ is con-s at a.

After we know $\frac{1}{g(x)}$ is con-s & $f(x)$ is also con-s, we can argue that $f(x) \cdot \frac{1}{g(x)}$ is also continuous as well using the result from (a).

$\Rightarrow \frac{f(x)}{g(x)}$ is continuous at $x = a$.

$\blacksquare$

Problem 2.1.6.

Use the definition of continuity to show that if $f(x) = \frac{1}{\sqrt{x}}$ is con-s at all points $a > 0$.

Solution:

To use the result from Problem 2.1.5.(b),

let's show $g(x) = \sqrt{x}$ is con-s, instead.

WTS:

(1) $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $|\sqrt{x} - \sqrt{a}| < \varepsilon$ whenever

$|x - a| < \delta$ for all $a > 0$.

To start with,

(2) $$|\sqrt{x} - \sqrt{a}| = \left| \frac{(\sqrt{x} - \sqrt{a})(\sqrt{x} + \sqrt{a})}{(\sqrt{x} + \sqrt{a})} \right| = \frac{|x - a|}{|\sqrt{x} + \sqrt{a}|} < \frac{\delta}{|\sqrt{x} + \sqrt{a}|}$$

(3) $\left[\sqrt{x} \geq 0 \text{ whenever } x \geq 0 \right]$. So,

(2) $$|\sqrt{x} - \sqrt{a}| < \frac{\delta}{|\sqrt{x} + \sqrt{a}|} < \frac{\delta}{|0 + \sqrt{a}|} = \frac{\delta}{\sqrt{a}} = \varepsilon \Rightarrow$$

(4) $\delta = \delta(\varepsilon) = \varepsilon \cdot \sqrt{a}$

So, $\forall \varepsilon > 0$, we can provide $\delta = \sqrt{a} \cdot \varepsilon$ s.t. $\sqrt{x}$ is con-s. at all $a > 0$.

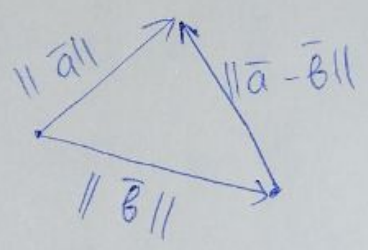
Using the Prop. 2.1.9, $\frac{1}{\sqrt{x}}$ is con-s at $a > 0$.

$\blacksquare$

2.1. Problem 7.

Prove $\big| \|\bar{a}\| - \|\bar{b}\| \big| \leq \|\bar{a} - \bar{b}\| \quad \forall a, b \in \mathbb{R}^n$.

Solution:



Using the Δ inequality

$$\begin{cases} \|\bar{a}\| + \|\bar{a} - \bar{b}\| \geq \|\bar{b}\| & (1) \\ \|\bar{b}\| + \|\bar{a} - \bar{b}\| \geq \|\bar{a}\| & (2) \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} \|\bar{a} - \bar{b}\| \geq \|\bar{b}\| - \|\bar{a}\| & (1) \\ \|\bar{a} - \bar{b}\| \geq \|\bar{a}\| - \|\bar{b}\| & (2) \end{cases} \quad \text{and} \Rightarrow$$

$$\|\bar{a} - \bar{b}\| \geq \big| \|\bar{a}\| - \|\bar{b}\| \big|$$

$\blacksquare$